AQ1.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
2.1 Level 1:

1 point

Suppose $A = \begin{bmatrix} 2 & 4 & 5 & 1 \\ 11 & -2 & 9 & -6 \\ -3 & 4 & 7 & 7 \end{bmatrix}$A = [211−34−245971−67]. Which of the following is true about the matrix AA?
[Hint: The (i, j)(i, j)-th entry is the entry which is at the ii-th row and jj-th column.]

☐ It is a $4 \times 3$4 × 3 matrix.

☑ It is a $3 \times 4$3 × 4 matrix.

☐ $(2, 3)$(2, 3)-th entry of the matrix AA is 4.

☑ $(2, 3)$(2, 3)-th entry of the matrix AA is 9.

Yes, the answer is correct.
Score: 1
Accepted Answers:
It is a $3 \times 4$3 × 4 matrix.
$(2, 3)$(2, 3)-th entry of the matrix AA is 9.

1 point
Which of the following statements is(are) TRUE?
[Hint: Recall, the definitions of scalar matrix, diagonal matrix, and identity matrix.]

○ Any diagonal matrix is a scalar matrix.
○ Scalar matrices may not be square matrices.
◉ Scalar matrices must be square matrices.
○ Any scalar matrix is an identity matrix.

Yes, the answer is correct.
Score: 1
Accepted Answers:
Scalar matrices must be square matrices.

1 point
Given below is the system of linear equations:

$$7x_1 + 10x_2 + 12x_3 = 36 \quad 8x_1 + 4x_2 - 9x_3 = 11 \quad 4x_1 - x_2 + 3x_3 = 10$$

7x1 + 10x2 + 12x3 = 368x1 + 4x2 − 9x3 = 114x1 − x2 + 3x3 = 10

If the matrix representation of the system of equations is $Ax = b$Ax = b, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = [x1x2x3], then

◉

$A = \begin{bmatrix} 7 & 10 & 12 \\ 8 & 4 & -9 \\ 4 & -1 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$A = [784104−112−93] , b = [361110]

○

$A = \begin{bmatrix} 7 & 10 & 12 & 36 \\ 8 & 4 & 9 & 11 \\ 4 & 1 & 3 & 10 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$A = [78410411293361110] , b = [361110]

○

$A = \begin{bmatrix} 7 & 10 & 12 & 36 \\ 8 & 4 & -9 & 11 \\ 4 & -1 & 3 & 10 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$A = [784104−112−93361110] , b = [361110]

○

$A = \begin{bmatrix} 7 & 10 & 36 & 12 \\ 8 & 4 & 11 & -9 \\ 4 & -1 & 10 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$A = [784104−136111012−93] , b = [361110]

Yes, the answer is correct.
Score: 1
Accepted Answers:

$A = \begin{bmatrix} 7 & 10 & 12 \\ 8 & 4 & -9 \\ 4 & -1 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$A = [784104−112−93] , b = [361110]

1 point

Which of the following statements is(are) TRUE?

- [ ] Addition of two matrices is possible only if the number of columns in the first matrix is same as the number of rows in the second matrix.
- [x] Addition of two matrices is possible only if the orders of both the matrices are the same.
- [ ] Defining ABAB is possible if the number of rows in the matrix AA is same as the number of columns in the matrix BB.
- [x] Defining ABAB is possible if the number of columns in the matrix AA is same as the number of rows in the matrix BB.

Yes, the answer is correct.
Score: 1
Accepted Answers:

Addition of two matrices is possible only if the orders of both the matrices are the same.
Defining ABAB is possible if the number of columns in the matrix AA is same as the number of rows in the matrix BB.

1 point

Suppose $P = \begin{bmatrix} 3 & -1 & 7 \\ 4 & 0 & 1 \\ 2 & -5 & 2 \end{bmatrix}$, $Q = [1 \; 4 \; -9]$, $R = [0 \; -3 \; 10]$, $D = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$

[Hint: If A is a matrix of order $m \times n$ and B is a matrix of order $n \times p$, then the order of AB is $m \times p$.]

- [x] The matrix PD is of order 3×1.
- [ ] The matrix PD is of order 1×3.
- [ ] The matrix QD is of order 3×3.
- [x] The matrix QD is of order 1×1.
- [x] The matrix DQ is of order 3×3.
- [ ] The matrix DQ is of order 1×1.
- [ ] The product QD is not defined.
- [x] The product QR is not defined.
- [x] The addition $P + Q$ is not defined.
- [x] The addition $P + D$ is not defined.

Yes, the answer is correct.
Score: 1
Accepted Answers:
The matrix PD is of order 3×1.
The matrix QD is of order 1×1.
The matrix DQ is of order 3×3.
The product QR is not defined.
The addition $P + Q$ is not defined.
The addition $P + D$ is not defined.

2.2 Level 2:

1 point
Choose the set of correct options.
[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and use the given conditions to find matrix A, for first two options.]

- [ ] If A is a square matrix of order 2 and $A^2 = I$, then $A = I$ or $A = -I$ where I is the identity matrix of order 2.

- [ ] If A is a square matrix of order 2 and $A^2 = 0$, then $A = 0$.
- [x] If A and B are square matrices of order 3 and $A + B = 0$, then $B = -A$.
- [x] If A is a scalar matrix of order 3, B is a non-zero square matrix of order 3 and $AB = 0$, then $A = 0$.

Yes, the answer is correct.
Score: 1
Accepted Answers:
If A and B are square matrices of order 3 and $A + B = 0$, then $B = -A$.
If A is a scalar matrix of order 3, B is a non-zero square matrix of order 3 and $AB = 0$, then $A = 0$.

1 point
If A is a square matrix of order 2 whose first column is denoted by C_1 and second column is denoted by C_2 (the left most column is taken as the first one) and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ then choose the set of correct options.
[Hint: Recall the definition of matrix multiplication in terms of rows and columns.]

- [x] The first column of AB is $b_{11}C_1 + b_{21}C_2$.
- [x] The second column of AB is $b_{12}C_1 + b_{22}C_2$.
- [ ] The first column of AB is $b_{21}C_1 + b_{11}C_2$.
- [ ] The second column of AB is $b_{22}C_1 + b_{21}C_2$.

Yes, the answer is correct.
Score: 1
Accepted Answers:
The first column of AB is $b_{11}C_1 + b_{21}C_2$.
The second column of AB is $b_{12}C_1 + b_{22}C_2$.

1 point
Choose the set of correct options.
[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and use the given condition to find matrix A for last two options.]

- [x] There exist some real matrices A and B, such that $AB = BA$.
- [ ] There do not exist any real matrices A and B, such that $AB = BA$.
- [ ] There does not exist any real matrix A, such that $A^2 = A$.
- [x] There exists some real 2×2 matrix A, such that $A^2 + A + I = 0$

Yes, the answer is correct.
Score: 1
Accepted Answers:
There exist some real matrices A and B, such that $AB = BA$.
There exists some real 2×2 matrix A, such that $A^2 + A + I = 0$

Suppose A is a 3×3 scalar matrix and (1,1)-th entry of the matrix A is 4. Suppose B is a 3×3 square matrix such that (i, j)-th entry is equal to $i^2 + j^2$. Find the (2,2)-th entry of the matrix $2A + B$2A + B.

16

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 16

1 point

Suppose $A = \begin{bmatrix}-1 & 2\\ -4 & 7\end{bmatrix}$A = [−1−427] and $A^2 - \alpha A + I = 0$A2 − αA + I = 0 for some $\alpha \in R$α ∈ R.

Find the value of αα.

6

Yes, the answer is correct.

Score: 1

Accepted Answers: